

Chapter 40 — Clinical and Urodynamic Evaluation of Lower Urinary Tract Dysfunction in Children

- **Evaluation is tailored to the individual child.** Presentations, underlying causes and phenotypes vary widely, and **age and timing of presentation** change what testing is needed.
- **Most lower urinary tract (LUT) pathology can now be diagnosed from history, examination and non-invasive tests**, avoiding invasive, complex and costly studies.
- **Testing serves four purposes:** diagnostic work-up, screening, surveillance, and follow-up after a procedure.

CLINICAL EVALUATION

History

- **Take the history from the child where possible**, pitched to age and to the maturation of voluntary bladder and bowel control.
- **Diagnostic sequence:** exclude an **anatomical or neurological cause** → if both are excluded, define the **type of bladder (and/or bowel) dysfunction**.
- **Components:** symptoms; voiding frequency; bowel function; previous urinary tract infections (UTIs); family history; maternal prenatal history, including screening ultrasonography; birth history; developmental milestones, including toilet training; neuropsychiatric conditions; past medical and surgical history; social history; diet; review of systems.

Storage Symptoms

- **Ask about urgency, frequency and incontinence.** Characterise incontinence as **continuous vs intermittent** and **day vs night**, and record how often it happens.
- **The history can suggest increased or decreased bladder capacity.** Confirm it with directed work-up.
- **Secondary enuresis** means new incontinence **after at least 6 months of dryness**. Explore new stressors at home or school.
- **Caffeinated drinks are bladder irritants** (soda, tea, coffee, sports and energy drinks) and directly cause urgency.

Voiding Symptoms

- **Males:** ask about a deviated or narrowed stream, prolonged voiding, incomplete emptying, straining, dysuria and post-void dribbling.
- **Females: toilet posture matters.** Sitting with the legs held together, slouching, perching, hovering, or dangling feet activates the pelvic floor. This **raises outlet resistance** → higher voiding pressures → dysuria and incomplete emptying → bladder dysfunction and UTI.
 - **Correct it with a footstool**, or by sitting "**backwards**" **facing the cistern**, which forces the legs apart.
- **Holding manoeuvres** (the "potty dance", leg-crossing, grabbing the perineum, squatting) postpone voiding against urge or **detrusor overactivity (DO)**.
 - Raised perineal pressure **inhibits detrusor contraction through the sacral reflex arc**, probably through the **gate control theory**: large-fibre afferent input suppresses small-fibre transmission in the dorsal horn.

- **The same mechanism is proposed for neuromodulation:** transcutaneous electrical nerve stimulation (TENS), posterior tibial nerve stimulation (PTNS) and sacral neuromodulation.

Concept — Posture and holding manoeuvres work through one lever, pelvic floor activation, in opposite directions. Deliberate activation suppresses the detrusor. Inadvertent activation on the toilet obstructs voiding. Treatment in both cases is control of the pelvic floor.

Bowel Function

- **Bowel and bladder dysfunction coexist through shared sacral innervation (S2–S4).** Bowel dysfunction may be the **sole cause** of the urinary symptoms.
- **"Is your child constipated?" is not enough.** Ask specifically about **stool frequency, faecal incontinence, abdominal pain, stool size and consistency.**
- **Bristol Stool Scale** (University of Bristol, 1997; the UK "Meyers scale"): seven forms, from **1 (separate hard pellets)** to **7 (watery)**. It is a communication aid for setting targets and **titrating fibre or laxatives.**
- **Target: a soft, smooth, daily stool passed without pain, Bristol 3–4.**
- **Rome IV (2016)** gives the consensus definition (Table 40.1). The definition of bowel dysfunction **remains controversial**, and the specific questions above usually suffice.

TABLE 40.1 ROME IV CRITERIA FOR FUNCTIONAL CONSTIPATION IN CHILDREN (Modified from Rasquin et al., 2006; Drossman, 2016)

	Functional constipation	Functional non-retentive faecal incontinence
Applies to	Developmental age ≥4 years , insufficient criteria for irritable bowel syndrome ; symptoms for at least 1 month	Developmental age ≥4 years
Rule	Two or more of the criteria below	All of the criteria below
Criteria	≤2 defecations per week; ≥1 faecal incontinence episode per week; retentive posturing or volitional retention; painful or hard stools; large rectal faecal mass; large stools that obstruct the toilet	Defecation in socially inappropriate places ≥ once a month; no inflammatory, anatomical, metabolic or neoplastic process; no faecal retention

- **What the table is for:** a symptom-based, **two-of-six** definition that makes "constipation" objective in the LUT clinic.
- **Pattern:** constipation is **retentive** (infrequent, hard, large stools; a rectal mass). Non-retentive faecal incontinence is **soiling without retention**.
- **Trap:** the two are **mutually exclusive on retention**. A child with a rectal faecal mass cannot have non-retentive faecal incontinence.

FIG. 40.3 ALGORITHM FOR FUNCTIONAL CONSTIPATION (FC) AND LOWER URINARY TRACT SYMPTOMS (LUTS) (Modified from Burgers et al., 2013, International Children's Continence Society)

- **Sequence:** history and examination → apply **Rome IV** → **7-day bowel diary (with Bristol score) + urinalysis (± culture) + pelvic ultrasound** → treat any positive finding.
- **Ultrasound triggers:**
 - **raised post-void residual (PVR)** → treat accordingly;
 - **rectal diameter >30 mm** → treat FC;
 - **bladder wall >6 mm empty and >3 mm full** → treat LUTS.

FIG. 40.3 BRANCHES — RESPONSE TO FIRST-LINE TREATMENT

Response	Next step
Both improve	Maintenance for at least 3–6 months

Response	Next step
FC improves, LUTS persist	Uroflow with electromyography (EMG) → escalate: urotherapy → anticholinergics → $\beta 3$ agonists → botulinum toxin → biofeedback → transcutaneous electrical neuromodulation
LUTS improve, FC persists	Check adherence $\pm$ intensify toilet training → escalate: polyethylene glycol (PEG) $\pm$ enemas → biofeedback → transcutaneous neuromodulation → if no improvement, paediatric gastroenterology referral (colonic transit, spinal magnetic resonance imaging (MRI), defecography, anorectal and colonic manometry)

- **Treat the bowel first and the two in parallel.** Whichever component persists decides the next investigation.

Clinical Aids

COMPARISON — BLADDER DIARY VS FREQUENCY-VOLUME CHART

	7-day bowel and bladder diary	48-hour frequency-volume chart
Best for	Elimination habits ; objective evidence of constipation	A primary complaint of voiding dysfunction
Records	Timing of voids and stools, Bristol type , urinary and faecal incontinence by day and night; removes recall bias	Volumes of intake and output , incontinence; measured with a "hat" or cylinder, ideally over a normal weekend ; days need not be consecutive , but each covers 24 hours
Yields	The bladder and bowel together	Functional or maximum capacity , polyuria , effect of drink type and volume (caffeine, fluids before bed)

- **A frequency-volume chart is mandatory in spontaneous voiders who fail first-line conservative therapy.** It is useful in any bowel and bladder dysfunction (BBD).
- **Diaries are cheap and simple, and they expose habits** that families do not notice.
- **Functional capacity is compared with expected bladder capacity (EBC):** infants ≤ 1 year = weight (kg) $\times$ 7 mL; older children = (age in years + 2) $\times$ 30 mL.

Urinary Tract Infections

- **Establish whether infections were febrile and/or symptomatic.** Asymptomatic bacteriuria is uncommon, and much rarer in boys.
- **A bagged specimen is unreliable unless negative.**
- **A febrile, culture-proven UTI suggests incomplete emptying or stasis.**
- **Separate non-specific LUTS from undocumented "UTIs"**, especially when cultures were negative. The family may need re-education.

Neurologic Function

- **Screen for occult neurological disease:** perineal numbness, change in gait, altered limb sensation, new problems with balance or coordination.
- **These are most worrying when new in a previously well child** presenting with **secondary enuresis** or new LUT problems.

Validated Questionnaires

- **Validated patient-reported outcomes objectively measure symptoms, quality of life and emotional impact.** The best are tested for reproducibility and validity in the target condition and population.

COMPARISON — PAEDIATRIC LUT QUESTIONNAIRES

Instrument	Domain	Items (scoring)	Max	Cut-off	Completed by	Distinguishing feature
Dysfunctional Voiding Symptom Score (DVSS), 2000	Symptoms: storage, voiding, posturing, bowel	10 (0–3, past month)	30	—	Child or parent	Adapted from the American Urological Association Symptom Index; best at detecting change with treatment
Dysfunctional Voiding and Incontinence Scoring System (DVISS)	Symptoms + 1 quality-of-life item	14 , mostly yes/no	35 + 3	9 (both sexes)	Child or parent	Best discrimination for diagnosis
Paediatric Urinary Incontinence Quality of Life Score (PinQ)	Quality of life	20 (0–4); internalising and externalising subscores	80	Higher = worse	Child; validated parent proxy	Unaffected by comorbidity; cross-cultural; does not track treatment response
Incontinence Symptom Index–Paediatric (ISI-P), 2007	Symptoms	11	44	9 (both sexes)	The child, aged 11–17	Adolescent self-report

- **Symptom scores (DVSS, DVISS, ISI-P) diagnose and track treatment. PinQ measures burden.** DVSS and PinQ are complementary.
- **Choosing an instrument. In a head-to-head comparison, DVISS discriminated diagnosis best; DVSS, DVISS and ISI-P all tracked subjective recovery; PinQ did not.**
 - **Why PinQ may not move:** therapy may not reach the domains it measures; emotional states are less malleable; emotional change lags behind symptom relief; or relief does not produce emotional change.
- **Psychological screening. Behavioural and emotional comorbidity is common** and rises from nocturnal enuresis → daytime incontinence → **faecal incontinence (highest)**. **Screen for it as routinely as organic causes are excluded.**

COMPARISON — CBCL VS SSIPPE

	Child Behavior Checklist (CBCL)	Short Screening Instrument for Psychological Problems in Enuresis (SSIPPE)
Role	Full assessment	Short screen
Content	100+ Likert items: internalising (anxiety, depression) and externalising (aggression, hyperactivity)	7 internalising items from the CBCL + 6 externalising items from the Attention Deficit Hyperactivity Disorder scale
Ages	18 months–18 years (versions for 18 months–5 years and 6–18 years)	Children with enuresis

- **Screening sequence:** SSIPPE → **≥2 positive items in either section** → full CBCL → disturbance confirmed → **child psychology referral**. Parents complete both.

Concept — A questionnaire answers only the question it was built for. Symptom scores diagnose and follow treatment. Quality-of-life and behaviour scores measure the burden on the child, not the response to treatment.

Physical Examination

- **Examine every child comprehensively**, with emphasis on **vital signs and the neurological, abdominal and genitourinary systems**.

Vital Signs

- **Record height, weight, temperature, heart rate and blood pressure at every visit.** Being **under- or overweight** is associated with higher LUTS scores.

COMPARISON — BMI VS TRI-PONDERAL MASS INDEX

	Body mass index (BMI)	Tri-ponderal mass index (TMI)
Formula	kg/m ²	kg/m ³
In children	Adult-validated (19th century); inaccurate unless converted to z-scores , which is cumbersome	Classifies healthy vs overweight better, with fewer false "overweight" labels
Thresholds	Percentile-based	Overweight ≥16.0 (M) / 16.8 (F); obese ≥18.8 (M) / 19.7 (F) kg/m³

- **Raised blood pressure** (on an **age-, sex- and height-specific nomogram**) may indicate **renal disease**, which often follows LUT pathology.
 - **Sequence:** repeat to confirm, ideally at separate visits → **renal and bladder ultrasound + urinalysis, serum creatinine and/or cystatin C**.

Neurologic Examination

- **Assess reflexes, strength, coordination and sensation in all limbs.**
 - Upper limb function says nothing about the LUT, but **fine motor control, sensation and handedness** decide whether **self-catheterisation is feasible**.
- **Document mobility aids** (wheelchair, walker, crutches, ankle-foot orthosis).
- **The Centers for Disease Control and Prevention (CDC) National Spina Bifida Patient Registry** records **functional neurological level** and **mobility** separately, because **they do not correlate**.

TABLE 40.2 OBJECTIVE CLASSIFICATION OF LOWER EXTREMITY FUNCTION IN TERMS OF FUNCTIONAL NEUROLOGIC LEVEL AND AMBULATION PHENOTYPE (AGE 2 OR OLDER) — (A) FUNCTIONAL NEUROLOGICAL LEVEL

Functional neurological level	Lowest preserved function
Thoracic	None: flaccid lower limbs
High lumbar	Hip flexion
Mid lumbar	Knee extension
Low lumbar	Foot dorsiflexion
Sacral	Foot plantar flexion

TABLE 40.2 (B) AMBULATORY PHENOTYPE OR MOBILITY STATUS

Ambulatory phenotype	Walking	Wheelchair
Perfect ambulator	Walks, runs and jumps everywhere, unassisted	None
Community ambulator	Indoors and outdoors for most activities; may use aids	Long trips out of the community
Household ambulator	Indoors only , with aids; transfers with little help	Some indoor activities; all community activities

Ambulatory phenotype	Walking	Wheelchair
Non-functional ambulator	Only in therapy sessions	At all other times
Non-ambulator	Never	Always; can transfer chair ↔ bed

- **What the table is for:** a standard vocabulary for spina bifida registries, from age 2 upwards.
- **Pattern:** level descends hip → knee → dorsiflexion → plantar flexion. Mobility steps down by **where** the child can walk.
- **Trap: level and mobility do not correlate.** A low lesion does not guarantee walking.
- **Assess deep tendon reflexes, gait, perineal sensation and anal tone.**

COMPARISON — SACRAL REFLEXES

	Anocutaneous reflex ("anal wink")	Bulbocavernosus (Osinski) reflex
Stimulus	Stroking the perianal skin	Squeezing the glans penis or clitoris
Response	External anal sphincter contracts	External anal sphincter contracts
Absent =	Interruption of the sacral reflex arc (S2–S4), which also controls bladder and bowel	Same

- **Inspect and palpate the spine for occult spinal dysraphism.** Most spines are normal.
 - Midline stigmata in healthy neonates rarely signify an occult lesion.
 - A simple sacral dimple has not been associated with dysraphism.

COMPARISON — SPINAL CUTANEOUS STIGMATA BY RISK

Risk	Cutaneous stigmata	Imaging
Low	Simple sacral dimple <2.5 cm from the anus	Spinal ultrasound, up to 6 months of age
Intermediate	Deviated gluteal cleft; dermal sinus tract	Not specified
High	Vascular malformation; lipoma or mass; hair tuft; multiple stigmata; VACTERL features (vertebral, anal, cardiac, tracheo-oesophageal, renal, limb)	MRI of the lumbar and sacral spine

COMPARISON — SPINAL ULTRASOUND VS MRI

	Spinal ultrasound	Spinal MRI
Use	Low-risk lesions	High-risk lesions
Age	≤6 months; the spine ossifies by 10–12 months	Any age
Trade-off	Cheap, avoids general anaesthesia; reduced sensitivity	Definitive; often needs anaesthesia

Abdominal Examination

- **Inspect** for asymmetry or distension (gas, faecal retention).
- **Palpate all four quadrants** for enlarged kidneys or bladder, masses, and **stool**. Stool is palpable **only in severe constipation**, more easily in small children.
- **Suprapubic fullness or tenderness** suggests **retention**.

Genital Examination

- **Look for damp underwear and genital irritation** as signs of incontinence.

COMPARISON — GENITAL FINDINGS BY SEX

	Male	Female
Look for	Circumcision status; balanitis; meatal stenosis	Urine pooling; inflammation; labial fusion; ectopic ureteric orifice; ventral meatal web
Key lesion	Meatal stenosis, which occurs in circumcised boys: irritation of the exposed meatus → epithelialisation → ventral web	Labial fusion: common <2 years, in nappies; rarely obstructs
When it matters	Only with significant stream deviation or narrowing. Counsel that concurrent symptoms may not be due to it or improve after surgery.	Pooling → stasis, colonisation, UTI
Management	Meatal inspection may suffice; the preferred method is to watch the stream	Labial fusion resolves by puberty as oestrogens rise; topical oestrogen if concern persists, about 10% recur
Continuous incontinence	—	Look for an ectopic ureteric orifice

- **Female meatal anomalies are associated with voiding dysfunction.** Not all are symptomatic, so **evidence for surgical correction is minimal.**

Concept — The examination hunts for what history cannot exclude: a neurological lesion (sacral reflexes, spinal stigmata, limb function), an anatomical outlet problem (meatus, labia, ectopic orifice) and retained stool. **The lower back is where occult neurology hides.**

KEY POINTS

- **History and examination establish whether the disorder is primarily one of storage or of emptying,** and work-up and treatment are tailored to that.
- **The complete history** covers symptoms, frequency, bowel function, UTIs, family, prenatal, birth and developmental history, neuropsychiatric, medical, surgical and social history, diet and review of systems.
- **Seven-day bowel and bladder diaries and 48-hour frequency-volume charts are invaluable.**
- **Questionnaires:** symptoms (**DVSS, DVISS, ISI-P**); quality of life (**PinQ**); behaviour (**CBCL, SSIPPE**).
- **Lower back: inspect** (dimples, deviated cleft, hair tufts, dermal sinuses, vascular malformations) and **palpate** (sacrum present, no mass).

LABORATORY TESTING**Urinalysis**

- **Dipstick urinalysis detects glycosuria, proteinuria, haematuria, leucocyturia and bacteriuria,** and records specific gravity.

URINALYSIS FINDINGS AND ACTIONS

Finding	Meaning	Action
Glycosuria	Diabetes → polyuria	Investigate for diabetes
Proteinuria	May be isolated	Repeat on a separate occasion or perform a 24-hour urine protein. Persistent → serum creatinine
White cells	Inflammation or infection	Interpret with symptoms

Finding	Meaning	Action
Microscopic haematuria	Common with infection; may occur alone in BBD	Isolated asymptomatic haematuria rarely needs work-up
Leucocyte esterase, nitrite, bacteria	Infection	Suspicious → culture and/or empirical antibiotics

- Perform urinalysis ± culture for irritative symptoms (dysuria, urgency, frequency).
- A negative urinalysis in LUTS is reassuring. The test is cheap and simple.

Urine Culture

- Do not culture routinely. It causes over-testing, false positives and over-treatment, especially in colonised bladders.
- Children on clean intermittent self-catheterisation (CIC) are at high risk of colonisation. Culture only with a high pre-test probability:
 - fever;
 - abdominal or suprapubic pain;
 - malaise;
 - new or worse incontinence;
 - new or worse bladder spasm.

Additional Testing

- Blood tests are rarely indicated. Use them only if kidney disease or its risk factors are suspected (e.g. persistent proteinuria → creatinine ± 24-hour protein).

KEY POINTS

- Children with LUTS should have urinalysis: specific gravity, protein, glucose, white cells, red cells, bacteria.
- Isolated malodorous urine lacks sensitivity and specificity for a positive culture.
- Blood testing is not recommended unless upper tract disease is suspected.

Concept — Laboratory testing is driven by pre-test probability. Urinalysis is universal because it is cheap and a negative result reassures. Culture and blood tests wait for symptoms or signs, because in colonised bladders a positive result without symptoms only leads to over-treatment.

URODYNAMIC EVALUATION

- Urodynamics spans non-invasive tests (uroflowmetry, pelvic ultrasound, patch EMG) and invasive cystometry. All need one vocabulary.
- The International Children's Continence Society (ICCS) standardised terminology (1998 → 2006 → 2014 update) codifies how results are described, so that studies compare like with like. It is essential reading.

Noninvasive Testing

Uroflowmetry

- Uroflowmetry measures flow over time as the child voids into a flowmeter. It is widely available.
 - Most important variables: voided volume, maximum flow rate (Qmax) and pattern. It also gives the average flow rate (Qavg).

- It can be combined with perineal patch EMG of the pelvic floor and external sphincter.

COMPARISON — ICCS UROFLOW PATTERNS

Pattern	Shape	Suggests
Bell	Smooth, symmetrical	Normal at any age, sex or volume
Tower	High, peaked, short	Overactive bladder: sudden, forceful detrusor contraction (not present in every case)
Staccato	Irregular, never reaches zero	Intermittent sphincter over-activity / incomplete pelvic floor relaxation; BBD
Interrupted	Irregular, with zero-flow segments	Underactive detrusor with Valsalva voiding, or detrusor-sphincter dyssynergia (DSD); EMG distinguishes them
Plateau	Flat, low, prolonged	Obstruction (outlet, posterior urethral valves, stricture) or an underactive bladder

- **Staccato vs interrupted: does flow reach zero?** Staccato = outlet activity during a continuous detrusor contraction. Interrupted = an inadequate detrusor, so flow stops between strains.
- **Tower vs plateau:** a strong detrusor against a free outlet vs normal effort against a resistant outlet.
- **Standardisation. Qmax is the most relevant variable. Record it only if the peak lasts ≥2 seconds**, because spikes are often artefact.
 - **The flow index** comes from a large healthy population across ages. It corrects for **total capacity (voided volume + PVR)**, because flow falls with **under- and over-distension**, and gives **one ratio of actual to expected flow**.
 - **It cannot classify staccato or interrupted curves.** It proposes smooth, staccato and interrupted as sub-types of three primary patterns.
- **Caveats.** A flow is interpretable only if:
 - the child is **toilet-trained**;
 - it is read as **emptying information only, with nothing about storage**;
 - **voided volume is ≥50% of EBC** (the flow index is valid down to **50 mL**; the pattern needs about **50–100 mL**);
 - **more than one curve** is obtained.

Concept — A flow curve's shape reflects the balance between detrusor and outlet. Too much detrusor gives a tower. Too much outlet gives a plateau. Outlet activity that comes and goes gives staccato; detrusor effort that comes and goes gives an interrupted curve. **EMG decides which side is at fault.**

Pelvic Ultrasound

- **Ultrasound is a mainstay**, either formal (renal and bladder) or **pre- and post-void in clinic**. It is quick, painless, **non-ionising**, and shows **bladder and rectum**.
- **It triages who needs invasive urodynamics.** In voiding dysfunction, enuresis and recurrent UTI, volume and wall measurements predict urodynamic abnormality.
- **In spina bifida, a thicker bladder wall correlates with higher pressure and lower compliance**, which threatens the upper tracts.

COMPARISON — PELVIC ULTRASOUND PARAMETERS

Parameter	Abnormal	Meaning
Bladder wall thickness	>3 mm full; >5 mm empty (>6 mm in the ICCS constipation algorithm, a different source study)	Increased detrusor workload; associated with voiding dysfunction, DO and pelvic floor over-activity. Thickness varies inversely with filling.
Bladder volume wall index (BVWI) (volume : wall ratio, % of expected)	<70% = thick-walled, small bladder; >130% = thin-walled, large bladder	A normal BVWI strongly predicts normal voiding. In enuresis, a normal BVWI predicts better response to desmopressin.
Rectal diameter	>30 mm without an urge to defecate	Bowel dysfunction / stercoral bolus; correlate clinically
Urinary debris	Present	Usually stasis; associated with a positive culture, specific but insensitive (evidence from largely asymptomatic screening cultures)

- *Post-void residual. Urine remaining after voiding or catheterisation.* It is **highly variable** within one individual and **distorted by over-distension**, so repeat it. **Abnormal PVR can be as low as >20 mL on repeat voids.**

POST-VOID RESIDUAL CUT-OFFS BY AGE

Age	Single PVR abnormal if... (bladder capacity = voided volume + PVR)
4–6 years	>30 mL or >21% of bladder capacity
7–12 years	>20 mL or >15% of bladder capacity

- **Valid only if the bladder is neither under-distended (<50% EBC) nor over-distended (>115% EBC) and PVR is measured <5 minutes after voiding.**
- **The allowance shrinks with age. A late or over-distended PVR is invalid, not abnormal.**

Patch Electromyography

- **Patch EMG is often paired with uroflow but is not universal.** Perineal or perianal electrodes record pelvic floor activity, which **should fall silent during voiding.**
- **Lag time = from the start of pelvic floor relaxation → the start of flow.**

EMG LAG TIME

Lag time	Meaning
<2 s	DO; urgency and frequency
2–6 s	Normal (full bladder)
>6 s	Primary bladder neck dysfunction

- **EMG shows non-invasively whether voiding is synergic or dyssynergic.**
- *Normal coordination (Fig. 40.9, after Blaivas, 1985).* Command to void → **sphincter EMG silent, sphincter opens → bladder neck opens** → detrusor pressure rises → flow. At "stop" → **the sphincter contracts** → flow ceases.

Other Measures

- **Functional bladder capacity** (total bladder capacity) is the **working day-to-day capacity.** It is taken as the largest diary void or voided volume + PVR. **Normal: about 70–115% of EBC (up to 130% in some studies).**

KEY POINTS

- **Non-invasive urodynamics** = uroflowmetry, patch EMG, pelvic ultrasound.
- **Uroflow** gives flow characteristics and pattern. **Qmax** is the most relevant quantitative variable.
- **Ultrasound** gives PVR, wall thickness, rectal diameter and debris.
- **EMG** shows synergic vs dyssynergic voiding, and gives lag time with uroflow.

Concept — *Non-invasive testing rebuilds the urodynamic picture piece by piece: outlet from uroflow; capacity, residual, wall and rectum from ultrasound; coordination from EMG. Each value is valid only under normative conditions, so the skill lies as much in rejecting an invalid study as in reading a valid one.*

Formal Urodynamics

- **Paediatric urodynamics** measures filling, compliance, capacity, pressures, DSD, DO and PVR. With fluoroscopy it also shows vesicoureteric reflux (VUR) and bladder, bladder neck and urethral anomalies.
- **There are no formal paediatric indications**, and practice varies. **Established uses:**
 - congenital or acquired neurological deficit;
 - severe obstructive uropathy;
 - possible neurogenic bladder, especially with new LUTS.

CONDITIONS REQUIRING FORMAL URODYNAMICS

Neurogenic causes	Severe obstructive causes
Spinal dysraphism, sacroccocygeal teratoma, sacral agenesis, transverse myelitis, spinal cord injury	Posterior urethral valves, urethral atresia, large ectopic ureterocele

- In non-neuropathic LUTS, urodynamics rarely justifies itself.
- **Uses:** baseline or surveillance; re-baselining before, and re-assessment after, medical or surgical therapy; ruling specific dysfunction in or out. **Congenital conditions change over time**, so studies are repeated.
- **Rule:** perform formal urodynamics only when a new or abnormal result will change management.

Unique Considerations in Pediatric Patients

- **Setting.** Experienced staff and a "kind, understanding and relaxed atmosphere" are essential, because many children arrive anxious. **Parents stay throughout; distract with television, games or tablets; standardise the technique.**
- **Expectations.** **Counsel the family beforehand.** Families often believe urodynamics will **confirm or exclude neurogenic bladder**. It describes function, not the underlying disease, and an office study may not reflect daily physiology.
- **Pain and anxiety.** Pain is usually mild, but a minority report moderate pain.
 - **Predictors:** high pre-test anxiety, sensory-motor alteration of the lower limbs, difficult catheterisation.
 - **Anxiety peaks at catheter placement.** With very difficult catheterisation, **place the catheters under general anaesthesia in theatre and study the child once awake.**
- **Infection screening.** Routine pre-study culture has low yield.
 - **Post-urodynamic UTI is rare (1.4%),** even though bacteriuria is common in at-risk children.
 - **Replacing routine cultures with targeted ones did not raise the infection rate.**

- **Risk factors: intermittent or overnight catheterisation, recent UTI, immunosuppression, symptoms on the day.**
- **Universal screening or prophylaxis → over-treatment. Select by new LUTS on or before the day.** Data on prophylaxis for high-risk children are inconclusive.
- **Sedation. Sedation may alter detrusor function** (general anaesthesia reduces contractions in animals).
 - **General anaesthesia is not recommended; avoid sedation.**
 - **If needed, sedate only for catheter insertion, done in advance.**
 - **During cystometry, ketamine is preferred to midazolam:** faster onset and recovery, better effect, possibly safer.
 - **One small, unrepeatable same-patient comparison found no change with sedation.**

Technique — the sequence.

1. **Empty the bladder first**, or drain it via the catheter.
2. **Lidocaine gel → bladder and rectal catheters**, using **the smallest size**, typically 6 Fr dual-lumen.
3. **EMG patches symmetrically at 3 and 9 o'clock perianally**, on skin prepared with **alcohol and fine abrasive paper**, and **secured**.
4. **Sample at >1000 Hz** to separate movement artefact from sphincter activity.
5. **Sitting vs supine makes no difference.**
6. **Use surface patches, not needle electrodes.** Needles are painful and give no better accuracy.

CATHETER ROUTES IN ALTERED ANATOMY

Situation	Vesical (Pves) catheter	Abdominal (Pabd) catheter
Vesicostomy	Foley, balloon inflated, on gentle traction to occlude	Rectal
Catheterisable channel, bladder neck open	Channel or native urethra	Rectal
Colostomy	Urethral	Via the colostomy
Females (option)	Urethral	Vaginal (comparable)

COMPARISON — CONVENTIONAL VS NATURAL-FILL CYSTOMETRY

	Conventional cystometry	Ambulatory (natural-fill) cystometry
Filling	Artificial infusion	Physiological diuresis
Environment	Laboratory	Natural; less Hawthorne effect and stress
Relative findings	Larger capacity, larger filling Pdet rise, lower voiding pressure	Smaller capacity, smaller filling Pdet rise, higher voiding pressure
Drawbacks	Artefact from fill rate	Time and effort; no standard paediatric protocol
Paediatric set-up	Urethral catheters	Suprapubic 6 Fr dual-lumen + rectal 5 Fr balloon catheter, placed about 24 h before

- **Filling.** EBC ranges from 30–50 mL (newborn) to 400–500 mL (young adult). Too fast a fill creates artefact, so the fill rate is scaled: 5–10% of EBC per minute.
- **Infusate between room and body temperature, never cold.** In a randomised trial, room vs body temperature did not change leak point pressures, voiding pressures or PVR. **Room temperature gave slightly higher capacity and pressure, which was not clinically significant.**

- **Stop filling at any of:**
 - **strong urge;**
 - **discomfort;**
 - **micturition;**
 - **leakage > fill rate;**
 - **Pdet consistently >40 cm H₂O;**
 - **volume >150% of EBC.**

COMPARISON — FIRST VS REPEAT FILLING CYCLES

Repeat fills	First fill	Later fills
Maximum Pdet, detrusor leak point pressure	Highest	Fall
DO contractions	Most	Fewer
Capacity, compliance, PVR	Comparable	Comparable
Presence of DO	Reproducible in most	Reproducible in most

- **ICCS changed its guidance:** it previously recommended **≥2 cycles**; since **2015** repeat cycles are **at the provider's discretion**.
- **Which fill to report is unspecified.** Interpreting the **first, usually worst, fill** is conservative and protects the upper tracts.
- **Differences between fills are attributed to bladder viscoelasticity.** Weigh the marginal gain against a longer test and more anxiety.

Video urodynamics. **Urodynamics + imaging, with contrast in place of saline.** It adds **bladder shape and trabeculation, VUR, diverticula, and bladder neck configuration (open vs closed), and times reflux or leakage against volume.**

COMPARISON — FLUOROSCOPIC VS CONTRAST-ENHANCED ULTRASOUND VIDEO URODYNAMICS

	Fluoroscopic	Contrast-enhanced ultrasound
Contrast	Radio-opaque	Ultrasound agent , compatible with the equipment
Radiation	Ionising	None
Status	Established	Recent innovation; strongly preferred by parents, judged more comfortable
Accuracy	No head-to-head comparison	—

Concept — Every paediatric modification serves one aim: making an artificial test reproduce the child's natural bladder. Fill slowly in proportion to EBC. Use warm infusate. Avoid sedation. Keep the child calm. Stop at physiological limits. Each artefact distorts the pressures the study exists to measure.

Interpretation of Studies

- **Interpretation belongs to a trained paediatric urologist, advanced practice provider or urodynamicist.**
- **Objective parameters are measured reliably, but subjective interpretation of paediatric studies agrees poorly, even between colleagues trained together. No standard reporting criteria exist.**
- **Urologic Management to Preserve Initial Renal Function (UMPIRE) study** (multicentre, myelomeningocele): **central review agreed with local risk categorisation in only about 50%.**
 - **Open questions:** leak vs void; when to stop; how far above EBC to fill; interpretation with **high-grade VUR.**

- **A modified Delphi consensus (North American experts, 2024) defined an eight-section report:**

1. indications
 2. study conditions
 3. procedural details
 4. filling phase
 5. radiological findings
 6. incontinence and leak point pressures
 7. voiding phase
 8. summary
9. It uses discrete options, calculated and conditional fields, and CPT (billing) codes, and is intended for **electronic records with decision support**.

Storage phase.

- **Pves** (bladder catheter) and **Pabd** (rectal catheter) are **zeroed to atmospheric pressure at the pubic symphysis**. $P_{det} = P_{ves} - P_{abd}$.
- **Normal filling: a flat Pdet, unchanged by cough, laugh or Valsalva.**

COMPARISON — DETRUSOR OVERACTIVITY VS UNDERACTIVITY

	Detrusor overactivity (DO)	Detrusor underactivity
Filling	Phasic Pdet rises; neurogenic vs idiopathic	Large capacity, low pressure, no contractions
Voiding	—	No sustained contraction; cannot empty; Valsalva voiding (large Pabd rises)
Pitfalls	Exclude provoked rises (cough, Valsalva); often unsensed; with an intact sphincter, guarding EMG rises	Not visible on Pdet; inferred from the combination of findings

- **Compliance.** ΔV (mL) / ΔP_{det} (cm H₂O), the ability to accommodate volume with little pressure rise. Calculate it in every patient. It cannot be inferred from Pdet alone.

COMPLIANCE GRADING

Compliance (mL/cm H ₂ O)	Grade
>20	Normal
10–20	Moderately impaired
<10	Severely impaired

- **Small, contracted bladders are a trap:** absolute Pdet may stay <30–40 cm H₂O while calculated compliance looks abnormal (<20). Here absolute pressure is the better guide to upper-tract safety.
- **An early Pdet rise = early loss of compliance:** the child spends most of the day at high pressure, which carries a worse prognosis.
- **Check the zeroing.** Errors hide poor compliance or give a falsely low Pdet.
- **Marked DO can make compliance unquantifiable.** The ICCS recommends using the most linear portion of the volume–pressure curve, ideally the whole fill.

COMPARISON — DETRUSOR VS ABDOMINAL LEAK POINT PRESSURE

	Detrusor leak point pressure (DLPP)	Abdominal leak point pressure (ALPP)
Leak occurs	Without an abdominal pressure rise or a detrusor contraction	With an abdominal pressure rise (cough, Valsalva), no detrusor contraction
Read from	Pdet	Traditionally Pves

	Detrusor leak point pressure (DLPP)	Abdominal leak point pressure (ALPP)
Reflects	The bladder's "pop-off" point	Outlet (bladder neck and sphincter) resistance
Emphasis	Upper-tract protection: the main paediatric measure	Outlet function only
Thresholds	<40 cm H ₂ O safe; >40, or no leak despite high pressure, = upper-tract risk	No paediatric norms. Adult female: <60 intrinsic sphincter deficiency (ISD); 60–90 equivocal; >90 normal
Bad result	High	Low

- **McGuire, 1981 (myelodysplasia, landmark):** a leak pressure >40 cm H₂O was followed by hydronephrosis or secondary VUR in 85%. 40 cm H₂O is the accepted upper limit, and the aim is the lowest achievable pressure.

Capacity.

- **Maximum cystometric capacity** = volume instilled, **confounded by heavy leakage**. Distinguish it from EBC (formula) and **functional capacity** (daily volume, **best from a diary**).
- **Abnormal capacity:** the ICCS defines **small** as <65% of EBC and **large** as >150%; some studies use >130%.
- **Abnormal capacity alone is not alarming.** It usually accompanies another finding: **large + infrequent voiding**, or **small + high Pdet** or **poor compliance**.

Voiding phase (pressure-flow study).

- **A pressure-flow study** = uroflow + pressures. It is valid only if the child voids ≥50–100 mL and ≥50% of functional capacity.
- **Record:** Qmax or Qavg, pattern, Pdet at Qmax, and PVR.
- **High Pdet + low Qmax = obstruction.**

COMPARISON — ANATOMICAL VS FUNCTIONAL OBSTRUCTION

	Anatomical obstruction	Functional obstruction (DSD)
Examples	Urethral valves, stricture	Detrusor contracts against a non-relaxing pelvic floor and sphincter
Coordination	Appropriate relaxation	No relaxation
Flow / Pdet	Plateau; high Pdet	Intermittent or staccato; Pdet higher than expected
EMG	Quiet	Active

- **To separate them, read flow, Pdet and EMG together.**
- **Normal Pdet at Qmax:** boys 66, girls 57 cm H₂O (similar to adults). With a catheter in situ, healthy **infant boys** reach up to 118 and **infant girls** up to 75 cm H₂O. High infant voiding pressures are physiological.

COMPARISON — FINDINGS THAT RAISE CONCERN FOR UPPER-TRACT INJURY

1. Impaired compliance (<10 severe; 10–20 moderate)
2. DSD with poor emptying, inability to void or high voiding pressure
3. Sustained raised Pdet during filling
4. DLPP >40 cm H₂O
5. High voiding pressure with poor flow

Concept — Interpretation is organised around one question: how much pressure do the kidneys see, and for how long? Compliance, DLPP, the timing of the pressure rise, DSD and voiding pressure all measure that exposure. **40 cm H₂O is the line**, because above it pressure is transmitted up the ureters.

Frequency of Testing

- The **UMPIRE protocol** (CDC National Spina Bifida Patient Registry, 2016) is a **research protocol, not a guideline**.
 - **Baseline at about 3 months** → repeat at **1, 2 and 3 years** → **yearly to 5** if there is **VUR** or a **"hostile" bladder**.

COMPARISON — ROUTINE VS EVENT-DRIVEN SURVEILLANCE

	Routine surveillance	Event-driven testing
Strategy	Annual urodynamics ± video	Test on clinical change : new UTI, new or worse incontinence or hydronephrosis
Argument for	Hydronephrosis is an insensitive marker . In a large spina bifida series, hydronephrosis improved with more drainage while poor compliance did not , and a minority needed surgery.	Cost ; many studies needed to detect an actionable change

- **Home bladder-pressure monitoring on CIC is investigational.**
- **Minimum consensus: regular follow-up of every neurogenic bladder**, for safe upper tracts, adequate emptying and low UTI risk. **Any change → investigate.**

KEY POINTS

- **Urodynamics is the gold standard** for evaluating LUT dysfunction.
- **Universal prophylaxis or culture before urodynamics does not prevent post-procedure UTI (1.4%).**
- **Paediatric considerations: sedation and fill rate scaled to EBC.**
- **Master:** capacity, Pves, Pabd, Pdet, compliance and **timing of its loss**, DLPP, ALPP, Pdet at Qmax, DSD, PVR.
- **DLPP and ALPP differ. DLPP dominates in paediatrics** because it relates to the upper tracts.

Concept — Surveillance trades test sensitivity against cost. Hydronephrosis is cheap but late, and urodynamics is sensitive but costly. That is why protocols front-load studies in infancy, when bladder behaviour is least predictable, and then space them out unless the bladder is hostile.

EVALUATION STRATEGIES FOR SPECIFIC CLINICAL ENTITIES

Non-neurogenic Lower Urinary Tract Dysfunction

- **This group includes:**
 - **BBD** (overactive bladder, underactive bladder, voiding postponement);
 - **non-monosymptomatic enuresis** (daytime symptoms + nocturnal enuresis);
 - **primary monosymptomatic nocturnal enuresis.**
- **Presentation:** urgency, frequency, day or night incontinence; storage- or emptying-predominant. **UTIs are more common and can confound the diagnosis.**
- **Most are diagnosed and treated without invasive tests:** history and examination → **selective non-invasive testing** → **behavioural and dietary change** → **medication** (anticholinergics, α -blockers, β 3-agonists).
- **Routine urodynamics is low-yield and rarely changes therapy.** **European Bladder Dysfunction Study (2008, prospective):** urodynamic findings did not correlate with treatment outcome.

- **Exception: failure of second- and third-line therapy** raises the chance of **actionable urodynamic findings**.

Spinal Dysraphism

- **Myelomeningocele = 90% of open spinal dysraphism**. These children need **lifelong study and aggressive treatment**, because LUT dynamics change with time.
- **Occult dysraphism (e.g. tethered cord) is harder**. Signs are subtle, and **the urinary tract may be the only system to change**.
- **Goals, whatever the cause:**
 1. **normal bladder volume;**
 2. **low storage pressure (normal compliance);**
 3. **complete emptying;**
 4. **upper-tract protection.**
- **Repeat urodynamics before any surgery** to confirm the planned procedure is correct.

UMPIRE REVISED RISK STRATIFICATION (SPINAL DYSRAPHISM)

UMPIRE revised risk	DLPP or end-fill pressure
Low	<25 cm H ₂ O
Intermediate	25–39 cm H ₂ O, or DO
High	≥40 cm H ₂ O

- **DSD was removed from the scheme** because the study team **could not agree who truly had it**. This reflects poor agreement, not that DSD is benign.

Sacral Agenesis

- **Partial or complete absence of the sacrum**, with abnormal nerve formation. **The radiographic level does not predict the clinical picture**.
- **Nearly one-third are diagnosed late**, with incontinence, failure to toilet-train or UTI.
- **Signs: flattened buttocks, short gluteal cleft, absent sacrum or coccyx on palpation**. Diagnosis is by **plain radiograph**, sometimes incidentally.
- **Urodynamics if voiding is dysfunctional**: either an **areflexic bladder with little sphincter function** or **DO with DSD**.

Anorectal Malformations

- **Anorectal malformations (including cloaca) are commonly associated with tethered cord**.
- **Tethered cord**: the **conus medullaris is fixed abnormally low (below its normal L1–L2 level)**. With axial growth, the cord comes under traction → gait change, altered leg and perineal sensation, **LUT dysfunction**.
- **Symptoms may be absent at birth**. Most recommend **baseline urodynamics even without a cord abnormality, especially after repair, at about 1 year after repair**. Any change → repeat the study.

Posterior Urethral Valves

- **Posterior urethral valves are the commonest cause of LUT obstruction in boys**. About 15% develop "valve bladder syndrome", with **progressive hydronephrosis and chronic kidney disease**.
- **Natural history**: **DO → loss of compliance → myogenic failure**.
- **Polyuria from nephrogenic diabetes insipidus** makes bladder emptying harder to optimise.

- **Dynamics change with time**, so follow-up is long-term; **clinical change → further study**.

Pelvic Tumors Requiring Extirpation

- **Resection of sacrococcygeal teratoma** (a rare neonatal germ cell tumour) causes **LUT dysfunction in up to 50%**, through **sacral plexus injury**.
- Symptoms may be hidden in the newborn period. **Keep a high index of suspicion**.
- Urodynamics shows **DO, DSD or areflexia**.

Cerebral Palsy

- **Voiding dysfunction affects up to one-third**. The lesion is **upper motor neuron**, and **storage symptoms outnumber voiding symptoms**.
- **Work-up is stepwise**: evaluation → conservative therapy → **urodynamics only if the response is poor**.
- **Findings**: **increased capacity**, **incontinence**, **under- or overactivity**, and **DSD only rarely**.
- **Spastic quadriplegia or moderate-to-severe impairment → higher risk of deterioration**, so monitor closely.

Spinal Cord Injury

- **Acute phase**: **spinal shock → retention → start a drainage programme**. The chronic phase shows **varied phenotypes**.
- **Higher lesions (cervical, thoracic) → more DSD** than lumbar lesions.
- **UK guidance**:
 - **baseline video urodynamics 3–6 months after injury**;
 - **surveillance** for deterioration, changed goals, or previously "**unsafe**" findings (e.g. poor compliance).

Transverse Myelitis and Other Central Nervous System Disorders

- An **inflammatory myelopathy** with the same **spinal-shock → chronic** course as spinal cord injury.
- **Motor recovery and bladder recovery are dissociated**. About half regain motor function, but **most retain voiding and bowel dysfunction**. Neither motor recovery nor spasticity predicts the bladder.
- **Initial urodynamics, commonest first**:
 1. **DO**
 2. **loss of compliance**
 3. **DSD**
 4. **DLPP >40 cm H₂O**
- **In early retention, preserved reflexes predict bladder recovery**.
- **Recommendation**: **early ultrasound and urodynamics + CIC** to protect the upper tracts.

COMPARISON — SPECIFIC CLINICAL ENTITIES

Entity	Mechanism	Typical urodynamics	When to study
Non-neurogenic LUTD	Functional	Rarely changes management	Only if refractory
Spinal dysraphism	Neural tube defect; tethering	Graded by DLPP or end-fill pressure	Baseline, serial, before surgery
Sacral agenesis	Absent sacral segments	Areflexia + weak sphincter, or DO + DSD	If voiding dysfunction

Entity	Mechanism	Typical urodynamics	When to study
Anorectal malformation	Associated tethered cord	Variable	Baseline about 1 year after repair; on change
Posterior urethral valves	Obstruction → remodelling	DO → poor compliance → myogenic failure	Serial; on change
Sacrocccygeal teratoma	Sacral plexus injury	DO, DSD, areflexia	Surveillance
Cerebral palsy	Upper motor neuron	Large capacity; DO or underactivity; DSD rare	Only if refractory
Spinal cord injury	Shock → chronic	Higher lesion → more DSD	Video urodynamics 3–6 months after injury; surveillance
Transverse myelitis	Inflammatory	DO > poor compliance > DSD > DLPP >40	Early, with CIC

- **Pattern:** urodynamics is **serial and mandatory** for **structural or progressive** lesions (dysraphism, tethering, valves, cord injury, myelitis). It is **reserved for treatment failure** in **functional or static** conditions (non-neurogenic LUTD, cerebral palsy).
- **Trap:** motor recovery or a normal newborn does not predict the bladder.

Concept — *The decision to perform urodynamics depends on whether the bladder faces a lesion that changes over time. Progressive or occult neurology and obstruction can damage the bladder silently, so they need a baseline and serial studies. Functional and static conditions are managed clinically, and urodynamics is kept for treatment failure.*